



Improvement in power inductor performance at 3 MHz by mixing carbonyl iron powder with Fe–Si–Cr crystalline alloy

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Abstract

We investigated the magnetic properties of Fe–Si–Cr crystalline alloy (FSC) by adding carbonyl iron powder (CIP) composites. Toroid core was prepared by mixing in various mix ratios FSC and CIP. Comparison of the density of the cores and the calculated packing fraction showed a similar tendency. Comparison of the packing fraction/permeability relationship and the Ollendorff equation proved that as the packing fraction increases, the permeability increases. The core-loss shows a decreasing tendency as the packing fraction increases. The sample, which shows the highest permeability and packing fraction, showed the lowest core loss at 3 MHz of 112.38 mW/cm³.

Introduction

Recently, soft magnetic materials are being used in various magnetic devices and power applications, such as chip inductors or transformers. Soft magnetic composite (SMC) characteristics showing high magnetic saturation (M_s), low coercivity (H_c), and low core loss (P_{cv}) are important for minimizing devices, while still maintaining high energy efficiency.^[1,2] Originally, because of their high permeability characteristic, spinel ferrites were widely used for power applications. However, because of their low magnetization saturation and low DC bias characteristic, these materials showed a lack of stability at the high-frequency range.

Much research has been conducted on soft magnetic materials to cope with challenging demands, such as permeability stability and high current stability in the high-frequency bands above 1 MHz, which are essential for power inductor application.^[3] Metallic soft magnetic materials show low hysteresis loss because of their low coercivity and show better performance on processability and stability at high frequency.^[4,5]

When fabricating SMCs, several important characteristics need to be determined. First, increasing permeability is important to achieve high inductance per volume. In this study, to increase the packing fraction that can increase permeability, we mixed relatively large size FSC and relatively small size CIP. Second, in order to disturb the eddy current flow, the use of materials with high resistivity is important. In our study, we used FSC, because they show relatively higher electrical resistivity caused by the Cr-rich oxidation layer on the particle's surface, which can result in reducing the eddy current loss.^[6,7] Because of its low cost, high magnetic saturation, and high complex permeability, CIP was used as the added smaller powder.^[8–10] Also, because of its large plastic deformation under

pressure during compaction to make the core, CIP is useful for increasing density.^[11]

Recently, SMCs mixed with insulating polymer have been proven to improve the overall permeability, and to prevent inter-particle eddy current loss (P_e), which reduces core loss (P_{cv}).^[12–14] In particular, bimodal cores composed of magnetic powders in various sizes are good candidates to increase permeability and magnetization conducted by high packing fraction.^[15,16]

To the best of our knowledge, there has been no report of the magnetic performances and core loss at high frequency at 1–3 MHz of FSC and CIP composite in various mixing ratios. Also, the relationship between packing fraction and permeability based on Ollendorff equation was not reported. In this study, we selected FSC as the major powder that was a few tens of micron in size; and as the minor powder, few micro-size CIP was chosen. We measured the microstructure, AC magnetic properties, and core loss of samples with various mixing ratios.

Methods

FSC and CIP were prepared by gas atomizing. The composition of FSC showed approximately 87, 11, and 2 wt% of Fe, Si, and Cr, respectively. FSC powders were distributed by sieves, and we selected the size of under 25 μm for experiment. FSC and CIP were mixed in weight ratios of 100:0 to 60:40. The powder mixture was blended with 3.0 wt% of binder containing insulating epoxy and dispersant. The blended mixture was mechanically mixed, and the mixture was pressed at a pressure of 1.5 ton/cm² at room temperature, in order to fabricate a toroidal core with inner diameter of 13 mm and outer diameter of 20 mm. The fabricated cores were hardened in a drier at 200°C

for 2 h under air. The sample made up of 100 wt.% of FSC was named Fe–Si–Cr in the figures. Weight ratios of 90:10, 80:20, 70:30, and 60:40 were named FC91, FC82, FC73, and FC64, respectively.

The characteristics of the raw FSC and CIP materials were measured by particle size analysis (PSA), X-ray diffraction (XRD) using Cu–K α , and scanning electron microscopy (SEM). The DC magnetic properties were measured using vibrating sample magnetometry (VSM). Impedance analysis was utilized to measure the permeability and Q-factor of the core. Core loss was measured by B–H analyzer over various frequencies. The fabricated cores were wound 10 times by Cu wire for the impedance analysis and B–H analysis measurements.

Results and discussion

Figure 1(a) shows the particle size distribution of both FSC and CIP. According to the data, FSC has an average size of approximately 20 μm , while CIP shows approximately 1.2 μm of average particle size. Figure 1(b) shows the XRD patterns. Cu–K α was used for the measurement, and it was taken in the range of 20–90° at intervals of 0.01°. The XRD peaks of FSC alloy and CIP were compared with the phases of α -Fe (ICDD #06-0696). Both powders showed the same main peaks corresponding to the α -Fe. In particular, the main peaks of CIP at 44.67°, 65.02°, and 82.33° show that the space group of CIP has $Im\bar{3}m$ symmetry.^[17] The narrowness and high intensity of the peak show that the samples are crystalline. Figure 1(c) shows the magnetic field-dependent magnetization of the prepared raw powders. The measured magnetization saturation and coercivity of FSC showed 157.07 emu/g and 6.9191 Oe, respectively.

The magnetization saturation (M_s) and coercivity (H_c) of CIP were measured as 190.59 emu/g and 4.1110 Oe, respectively. The VSM data reveal that the prepared raw powders show soft magnetic properties with low H_c . Low H_c induces low hysteresis loss, which can affect total core loss.^[18] Also, high M_s can induce good DC bias characteristic, which means that as the current increases, the inductance of the material is sustainably maintained.^[19]

Figure 2 shows the morphology of the FSC alloy and CIP that was investigated by scanning electron microscopy (SEM). The SEM imagery reveals that both powders show a uniform spherical shape because of gas atomization fabrication and confirms the result of the particle size data obtained by PSA. Based on the particle packing theory, the maximum density can be obtained through a cubic body having one-sized spherical shape particle perfectly layering in the body-centered cubic.^[20] Suzuki et al.^[21] suggested increasing the density of bulk body that consisted of particles by diminishing the porosity. Based on our previous research, the addition of CIP, which is the minor particle, into FSC alloy, the major particle, can be such a method of increasing the density.

Figure 3(a) shows the relationship between the measured density and the calculated packing fraction of the bulk of cores. The density of cores was measured by simple calculation by measuring the inner and outer diameters, and the height of the core for volume calculation, divided into the mass of the core. The packing fraction was calculated by the equation [$d_{\text{core}}/d_{\text{bulk}} \times 100\%$].^[22] The density of the core gradually increases from 4.65 to 5.43 g/cm³ for FSC to FC73. However, it decreases slightly at FC64, which shows density of 5.27 g/cm³. The tendency of the packing fraction is similar to that of

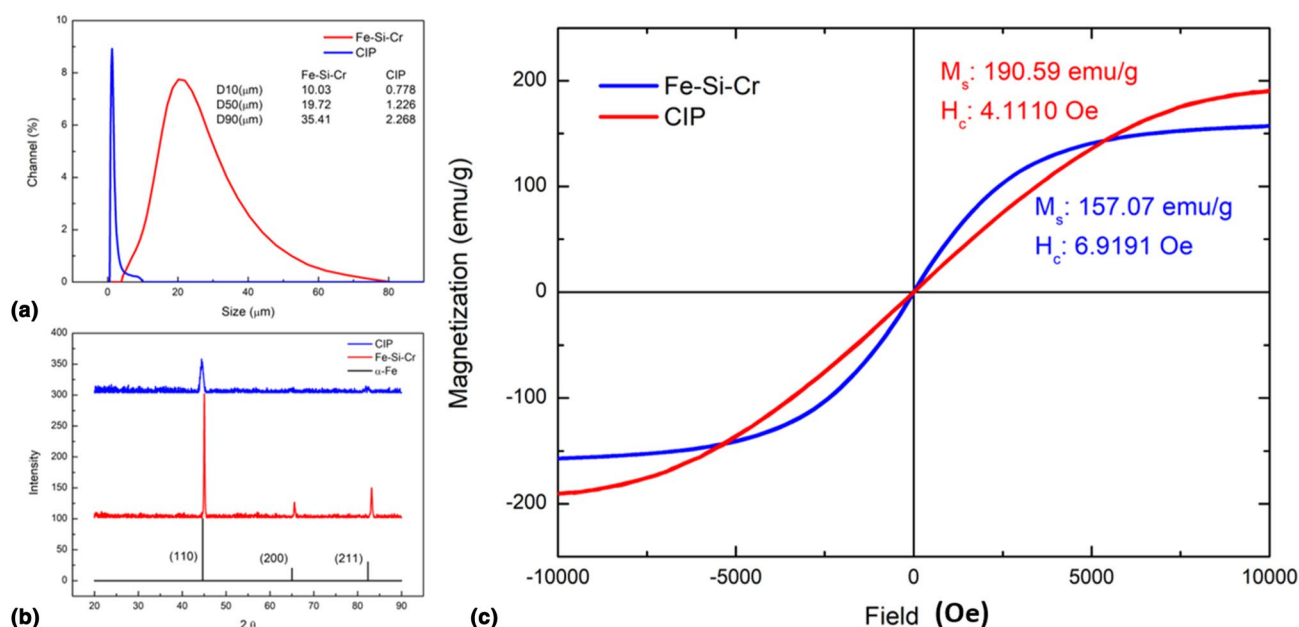


Figure 1. (a) Particle size distribution of each Fe–Si–Cr alloy and CIP. (b) XRD pattern of Fe–Si–Cr alloy and CIP compared to α -Fe based on ICDD card information. (c) magnetic hysteresis curve of Fe–Si–Cr alloy and CIP analyzed by VSM.

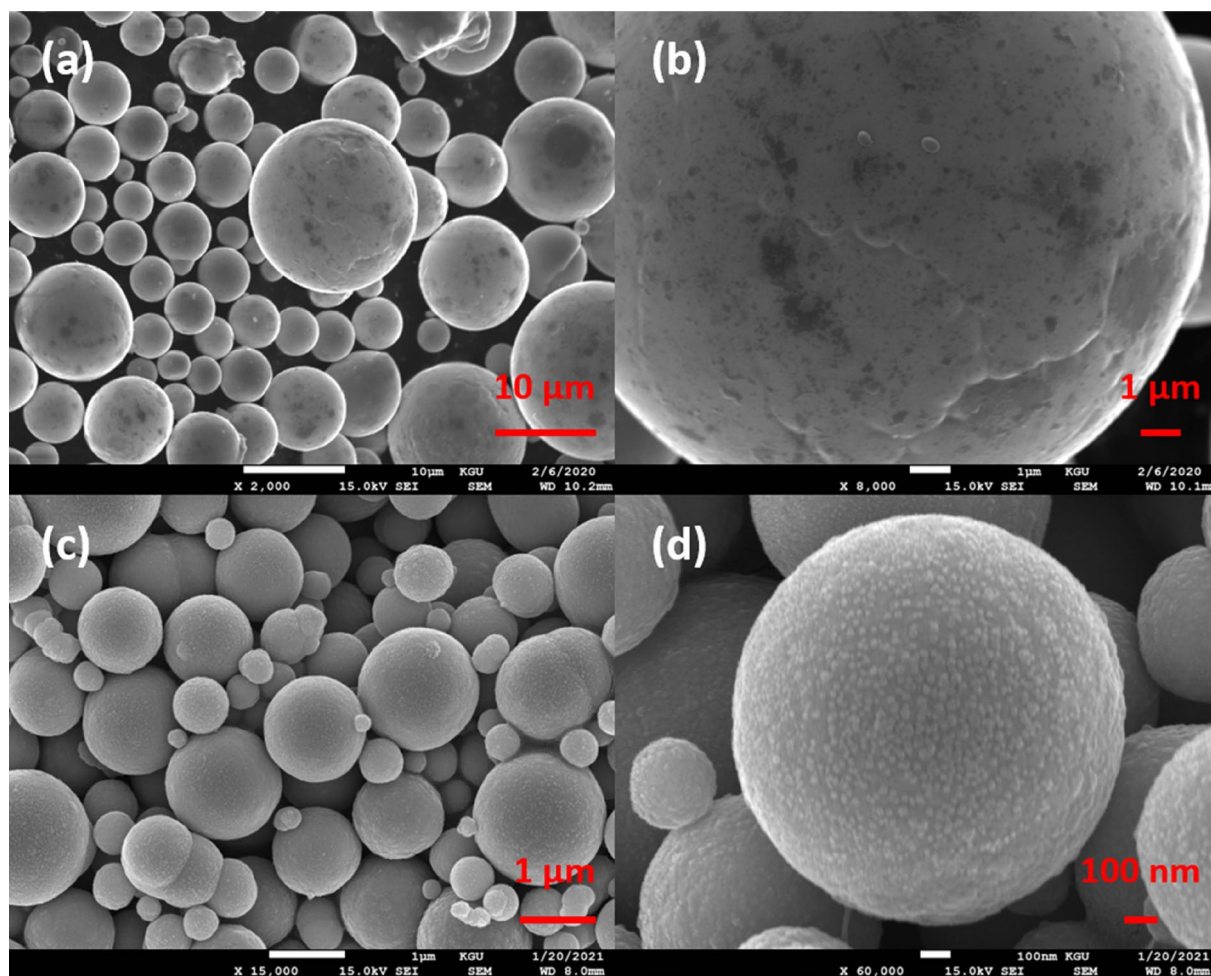


Figure 2. Morphology of (a), (b) Fe-Si-Cr alloy and (c), (d) CIP observed by SEM.

the core density, which shows an increase from FSC to FC73 of 64.67% to 73.47%, and slight decrease at FC64 of 70.67%. These behaviors result from the CIP, the minor particle, being packed between FSC particles of a few tens of microns in size. Figure 3(b) shows the magnetic hysteresis curve of the fabricated cores. Figure 3(c) shows that as the content of CIP increases, M_s gradually increases. This occurs because the M_s of CIP raw powder is higher than that of FSC expressed in Fig. 4. As the content of CIP increases, H_c gradually decreases, even though the coercivity of CIP is higher than that of FSC. This is because of the relationship between the packing fraction and coercivity, based on the study by Knowles.^[23] As the packing fraction increases, the coercivity decreases, except in the case of FC64. Permeability is the most important characteristic of power inductors because high permeability materials can induce high efficiency in power applications. Figure 3(d) shows the frequency dependency of permeability from 1 kHz to 110 MHz. Figure 3(d) reveals that all samples show stable permeability up to almost 20 MHz. The highest permeability was that of the FC73 sample, which also showed the highest density and

packing fraction. It includes the permeability at 1 MHz, which is the working frequency mainly used for various portable electronic devices.^[16] The tendency of permeability at 1 MHz showed similarity to the graph of the density and packing fraction relationship, which gradually increases as the content of CIP increases, and at CIP content of 40 wt%, it slightly decreases.

Equation (1) shows the Ollendorff equation. It was used to explain the relationship between packing fraction and permeability.^[24] The equation shows that the permeability of core is characterized by three main factors: η (volume packing fraction), N (coefficient of demagnetization factor, $0 \leq N \leq 1$), and μ_m (permeability of powder). The μ_m is the permeability in vacuum, with the value $\mu_m = 4\pi \times 10^{-7}$ H/m. In our work, we used $N = \frac{1}{3} \exp(-3\eta)$ for calculation based on Anhalt et al.^[25] The bulk permeability of Fe-Si-Cr alloy powder, 61 H/m, was referenced from Pan et al.^[26] Figure 4(a) compares the permeability between the computed Ollendorff equation and our fabricated samples, depending on the packing fraction. The tendency of our experimental data showed similarities to the equation-fitted data.

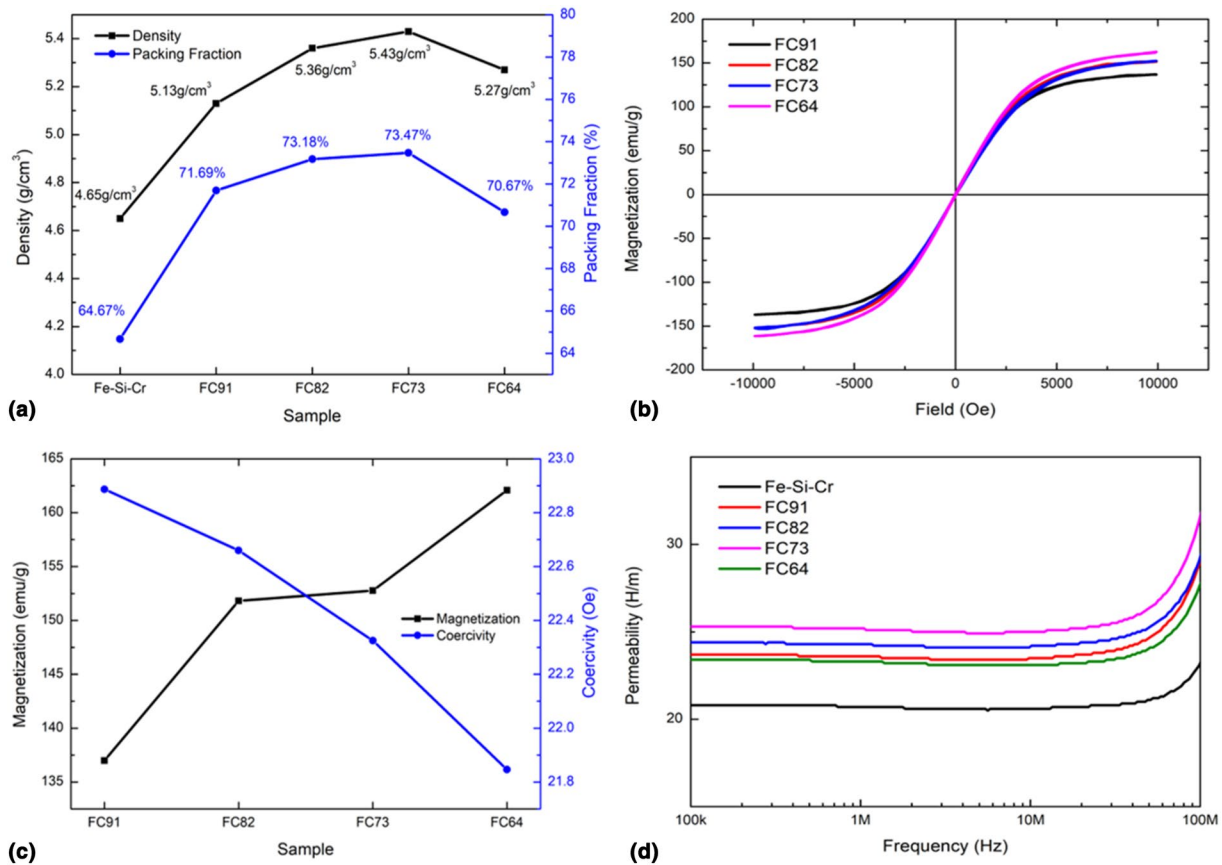


Figure 3. (a) Density and calculated packing fraction of each fabricated cores, (b) magnetic hysteresis curve of FSC + CIP composite by each composition, (c) magnetization and coercivity of each composition, and (d) the permeability as a function of applied frequency from 1 kHz to 110 MHz.

$$\mu = \mu_0 + \frac{\eta \mu_0 (\mu_m - \mu_0)}{N(1 - \eta)(\mu_m - \mu_0) + \mu_0} \quad (1)$$

Core loss measurement is also required for evaluating core materials applied to power applications. Figure 4(b) and (d) show the frequency dependence of core loss. Figure 4(b) was obtained under 20 mT flux density, measuring the frequency until 1 MHz. Figure 4(d) was obtained under 5 mT flux density to measure at higher frequency from 1 to 3 MHz. Figure 4(c) and (e) show the eddy current loss at 100 kHz~1 MHz and 1 MHz~3 MHz, respectively. Eddy current loss was measured by separating it from the total core loss. We used the curve fitting method in order to separate them. Figure 4(f) shows the core loss and eddy current loss of all samples, especially at 3 MHz. As the driven frequency increases, the core loss increases logarithmically. For total core loss and eddy current loss at 3 MHz, we obtained values of 112.38 mW/cm³ and 48.6165 mW/cm³, respectively, which were considerably low for the FC73 sample, representing the highest permeability. It was worth noting that the addition of CIP resulted in a reduction of total core loss compared to the use of only one metallic powder, FSC:

$$P_{\text{tot}} = P_h + P_e + P_r \cong P_h + P_e \cong W_h f + W_e f^2 \quad (2)$$

where P_h was the hysteresis loss, P_e was the eddy current loss, P_r was the residual loss, W_h was the hysteresis loss constant, and W_e was the eddy current loss constant. Core loss can be represented as the sum of the hysteresis loss, eddy current loss, and residual loss. In the case of magnetic metal cores, the residual loss can be ignored, because the value is exceptionally low.^[27] In our study, the M_s and H_c of composite samples did not show dramatic change, resulting in not much difference in hysteresis loss between the samples. As a result, eddy current loss is the main factor influencing the total core loss of composites. Eddy current loss can be divided into inter-particle eddy current loss and intra-particle eddy current loss. Inter-particle eddy current loss is a heat energy loss caused by the eddy current loss between the particles. Intra-particle eddy current loss is a heat energy loss caused by the eddy current loss inside the particles. In our study, a suitable amount of epoxy resin was used as an insulating binder, in order to decrease the inter-particle eddy current loss. Reducing the space that eddy current can flow is one of the methods to reduce intra-particle eddy current loss.^[28] This proves the result that the sample with the highest packing fraction and core density shows the lowest total core loss.

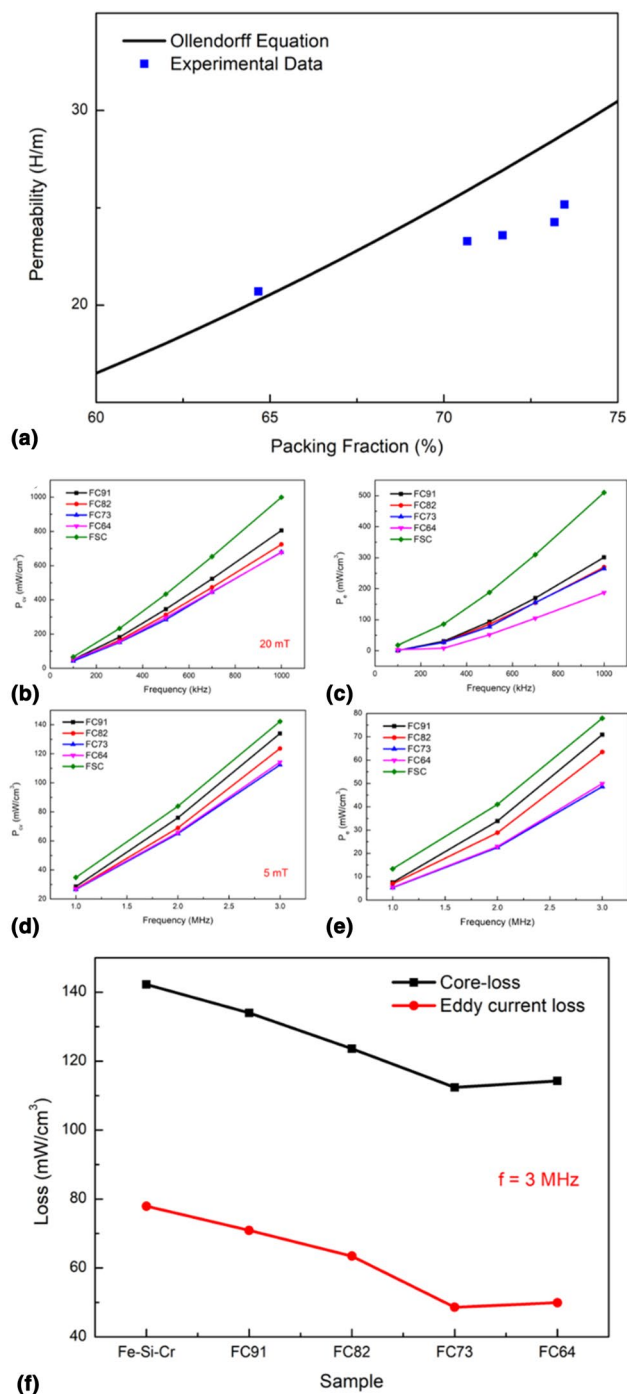


Figure 4. (a) Analysis of relationship between permeability to packing density, (b) core loss of each sample measured until 1 MHz under 20 mT, (c) eddy current loss of each sample until 1 MHz, (d) core loss of each sample measured from 1 to 3 MHz under 5 mT, (e) eddy current loss of each sample from 1 to 3 MHz, and (f) core loss and eddy current loss of samples at 3 MHz under 5 mT.

Conclusion

In this study, we investigated the AC magnetic properties of SMCs with various mixing ratios of FSC and CIP in the MHz

range. The core with mixing ratio 7:3 showed the maximum packing fraction of 73.47%. The core with maximum packing fraction also showed the highest permeability of 25.16 H/m at 1 MHz. Additionally, the core loss of the mixing ratio 7:3 sample showed 112.38 mW/cm³ at 3 MHz under 5 mT, which, in comparison to the other samples we fabricated, was the lowest core loss. The analysis of the core's permeability calculated by the Ollendorff equation showed that the permeability is strongly related to the core packing fraction. These experimental results show that the packing fraction, representing the intrinsic properties of the raw powder, as well as the microstructure of the cores, affects the core's characteristics for power application. They also indicate that the maximum packing fraction can result in the maximum permeability and minimum total core loss.

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